

## Question 15 – Stage 1

The curve  $\log_2(x) = \log_3(y)$  passes through the points  $(1, 1)$  and  $(8, 27)$ .

Let  $A$  be the area of the region between the curve, the  $x$ -axis and the lines  $x = 1$  and  $x = 8$ .

Let  $B$  be the area of the region between the curve, the  $y$ -axis and the lines  $y = 1$  and  $y = 27$ .

What is the value of  $|A - B|$  to 3d.p.?

### Working

A

$$y = 3^{\log_2(x)}$$

$$\int_1^8 (3^{\log_2(x)}) \, dx$$

```
In [5]: import scipy.integrate as integrate
from math import log

result = integrate.quad(lambda x: 3**(log(x,2)),1,8)
resultA = result[0]
errorA = result[1]

print(f"Error of A: {errorA}")
print(f"Result of A: {round(resultA,3)}")
```

Error of A: 9.871966057902894e-11  
Result of A: 83.173

B

$$x = 2^{\log_3(y)}$$

$$\int_1^{27} (2^{\log_3(y)}) \, dy$$

```
In [6]: result = integrate.quad(lambda y: 2**(log(y,3)),1,27)
resultB = result[0]
errorB = result[1]

print(f"Error of B: {errorB}")
print(f"Result of B: {round(resultB,3)}")
```

Error of B: 9.780016184346646e-07  
Result of B: 131.827

### Combining

```
In [7]: totalError = errorA + errorB
total = abs(resultA - resultB)

print(f"Total Error: {totalError}")
print(f"abs(A-B): {round(total,3)}")
```

Total Error: 9.781003380952437e-07  
abs(A-B): 48.653